

WIRED IN

Issue 04, 2026

**UNSW ELSOC**
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How can your degree lead to Telecommunications and Signals? This edition dives into the different skills and jobs in this specialty. See how you can make an impact in the world. Swipe to learn more.



EET Subfield: Telecommunications and Signals





WHAT IS TELECOM ENGINEERING?

Telecom engineering involves communication information at a distance - from mobile networks, Wi-Fi, satellites fibre optics and real-time wireless signals.

It is associated with data communication, encryption and compression stemming from the growing importance of digital and wireless networks.

Telcom Engineering involves designing, building and managing systems that transmit data, broadband signals and wireless

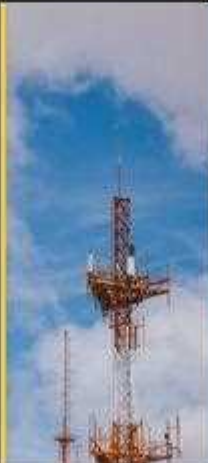
WHAT DO TELECOM ENGINEERS DO?

WIRELESS AND NETWORK SYSTEMS

Telecommunication Engineers design, build and maintain wireless and network systems that transmit signals over distances.

These include, mobile communications, IoT systems, satellite communications.

They work to improve speed, reliability and coverage while also testing



WHAT DO TELECOM ENGINEERS DO?

SIGNAL AND DATA PROCESSING

Telecommunication engineers work with signals and data so information can be transmitted efficiently without interference.

Engineers need to analyse, compress and process audio, video and wireless signals for applications such as streaming.

They help improve signal



WHAT DO TELECOM ENGINEERS DO?

DEVELOP AND RESEARCH TECHNOLOGY

Telecommunication engineers help develop and research new communication technologies that improve speed and security.

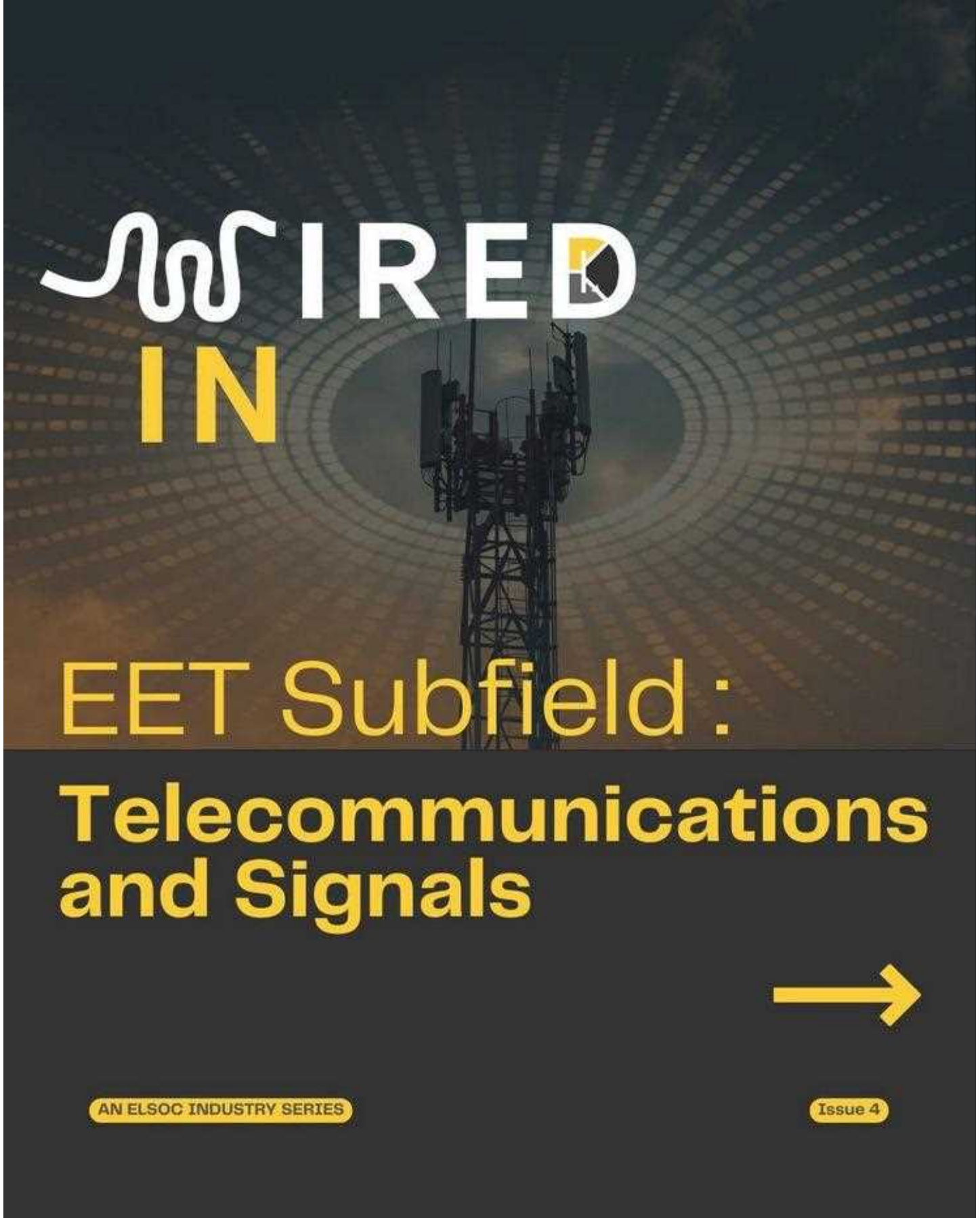
This includes areas such as 5G/6G, MIMO technology, terahertz communication, quantum communications and embedded systems which are rapidly growing research and industry



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Issue 4

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performance connectivity,



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SIGNAL AND DATA PROCESSING

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Engineers need to analyse, compress and process audio, video and wireless signals for applications such as streaming.

They help improve signal quality, reduce noise and manage data flow.

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What DO TELECOM ENGINEERS NEED TO KNOW

Key areas in EET include:



Electric Circuit Fundamentals (ELEC1111 + ELEC2134 + ELEC2141)



Electromagnetic Engineering (ELEC3115)



Digital And Analogue Signal Processing (ELEC3104 + TELE3113)



Network Security and Privacy (TELE3118 + TELE4641)



WHY PICK TELECOM AND SIGNALS ENGINEERING?



**Massive real world
impact through
global connectivity**



**Constant innovation
and expansion in
digital technology**



**Work with emerging
technologies like 6G
and quantum
communication**



**High demand
across software
and tech industries**

KEY TELECOMMUNICATION AND SIGNAL COMPANIES



Telstra, Optus (Wireless mobile communication)



NBN, Superloop (Fibre optic, broadband)



Ericsson, Nokia (5G equipment, network systems)



CSIRO (Telecommunications Research)



Aurecon, Arup, Jacobs (Consultancy)

